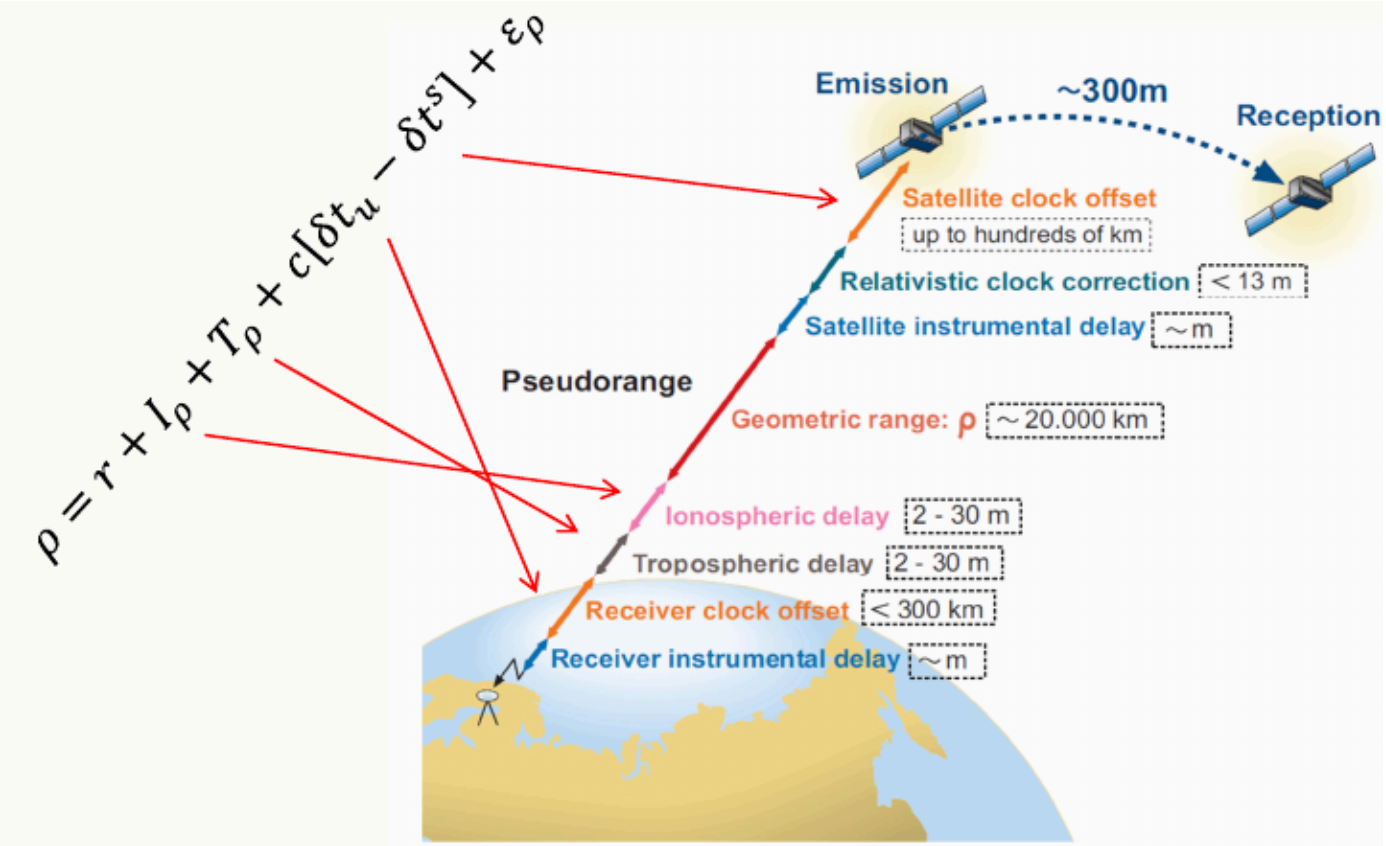
Lecture 6: Relative GNSS Using Carrier Phases



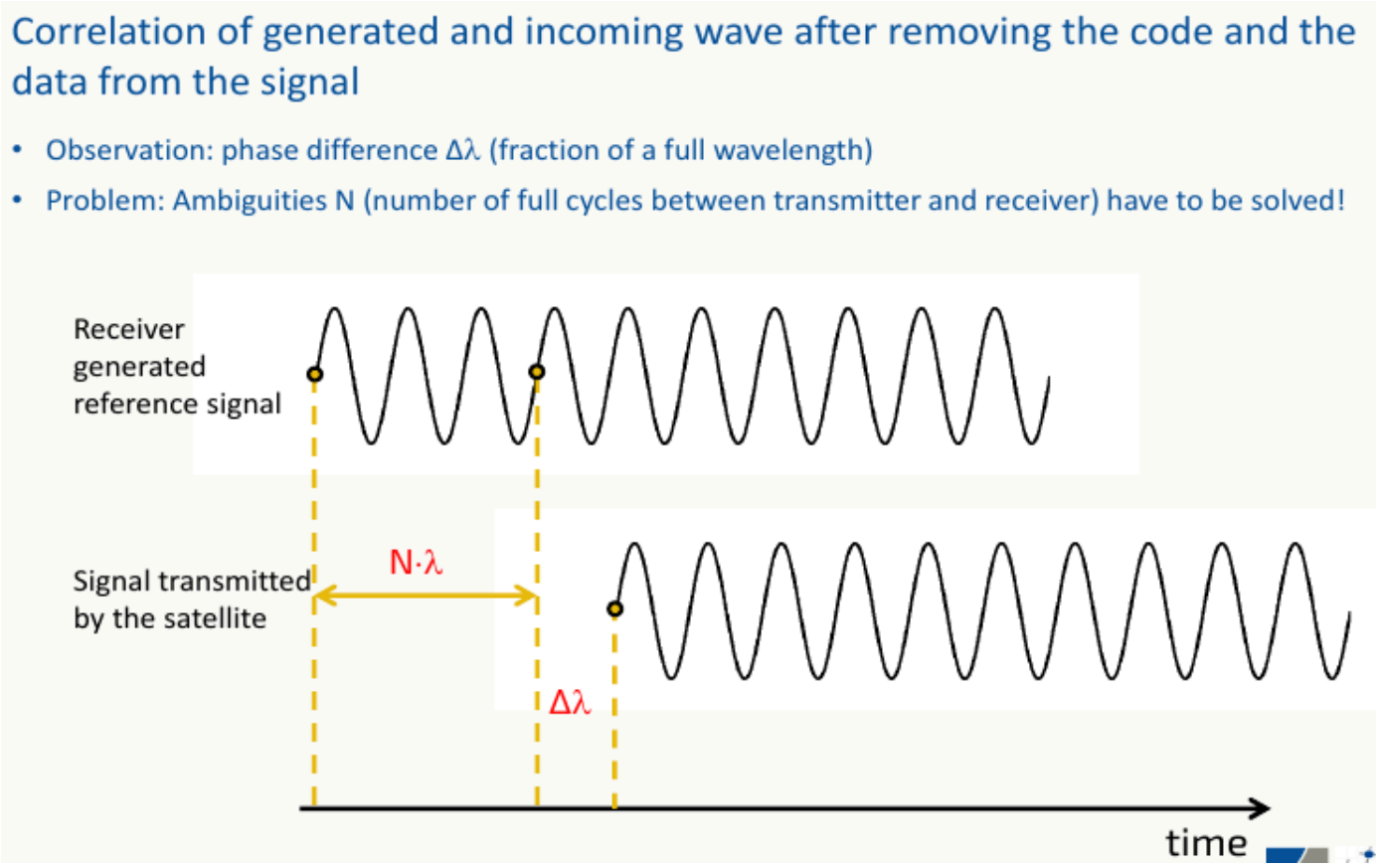
Code-Solution

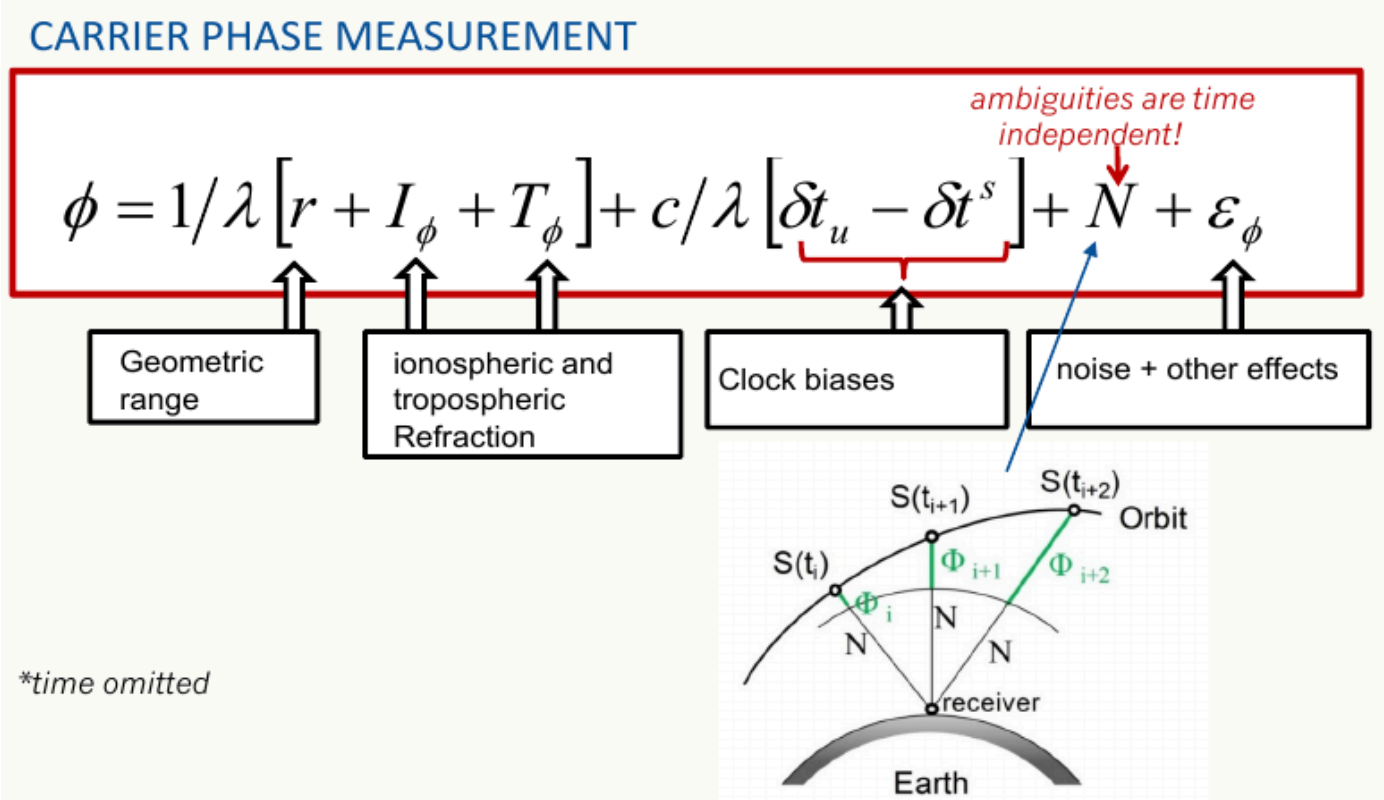
1. Correct for some errors
   - Calculate satellite clock offset (nav message)
   - Caluclate Ionospheric delay (nav message)
   - Calculate tropospheric delay (from height)
   - -> **Remaining errors: ~m**
2. Create observation model with corrected observations (relation between unknown parameters and observations)
   - $\rho_c^{(k)} = ||\boldsymbol{x}^{(k)} - \boldsymbol{x}|| + b + \tilde{\varepsilon}_\rho^{(k)}$
3. Approximation of non-linear model
   - $\boldsymbol{x} = \boldsymbol{x}_0 + \delta\boldsymbol{x} \quad b = b_0 + \delta b \quad \rho_c^{(k)} = \rho_0^{(k)} + \delta\rho^{(k)}$
4. Linearisation

$$\delta\boldsymbol{\rho} = \begin{bmatrix} \delta\rho_c^{(1)} \\ \delta\rho_c^{(2)} \\ \dots \\ \delta\rho_c^{(K)} \end{bmatrix} = \begin{bmatrix} (-\mathbf{1}^{(1)})^T & 1 \\ (-\mathbf{1}^{(2)})^T & 1 \\ \dots & \\ (-\mathbf{1}^{(K)})^T & 1 \end{bmatrix} = \begin{bmatrix} \delta\boldsymbol{x} \\ \delta b \end{bmatrix} + \boldsymbol{\varepsilon}_\rho = \boldsymbol{G} \begin{bmatrix} \delta\boldsymbol{x} \\ \delta b \end{bmatrix} + \boldsymbol{\varepsilon}_\rho$$

5. Solve iteratively using weighted squares
   - $\begin{bmatrix} \delta\boldsymbol{x} \\ \delta b \end{bmatrix} = (\boldsymbol{G}^{\boldsymbol{T}}\boldsymbol{P}\boldsymbol{G})^{-1}\boldsymbol{G}^{\boldsymbol{T}}\boldsymbol{P}\delta\rho$
   - $\boldsymbol{P} = \boldsymbol{\Sigma}_{ll}^{-1} = \frac{1}{\sigma_{URE}^2} 1_3$
   - $\boldsymbol{\Sigma}_{xx} = (\boldsymbol{G}^{\boldsymbol{T}}\boldsymbol{P}\boldsymbol{G})^{-1}$

Carrier Phase Measurements

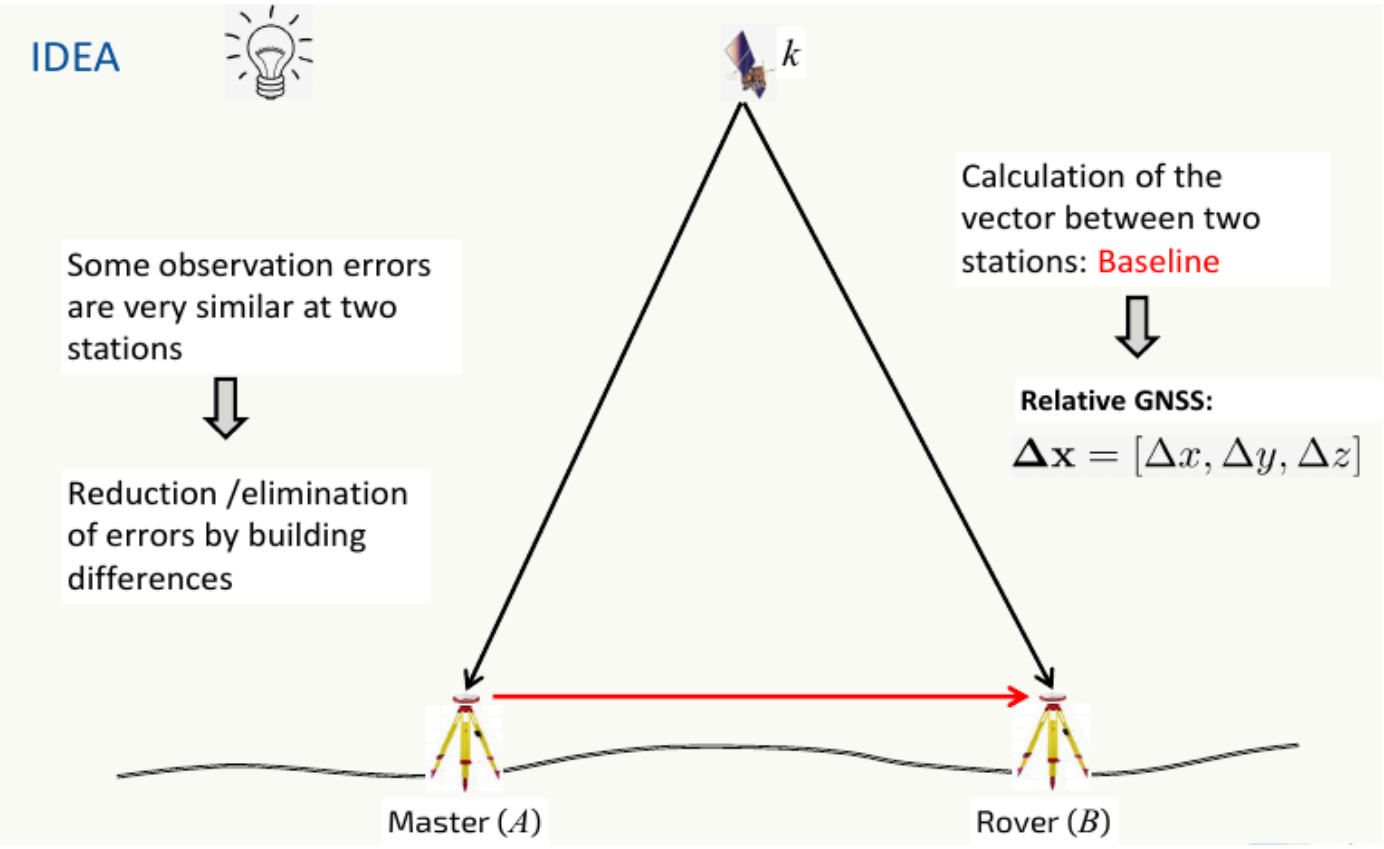


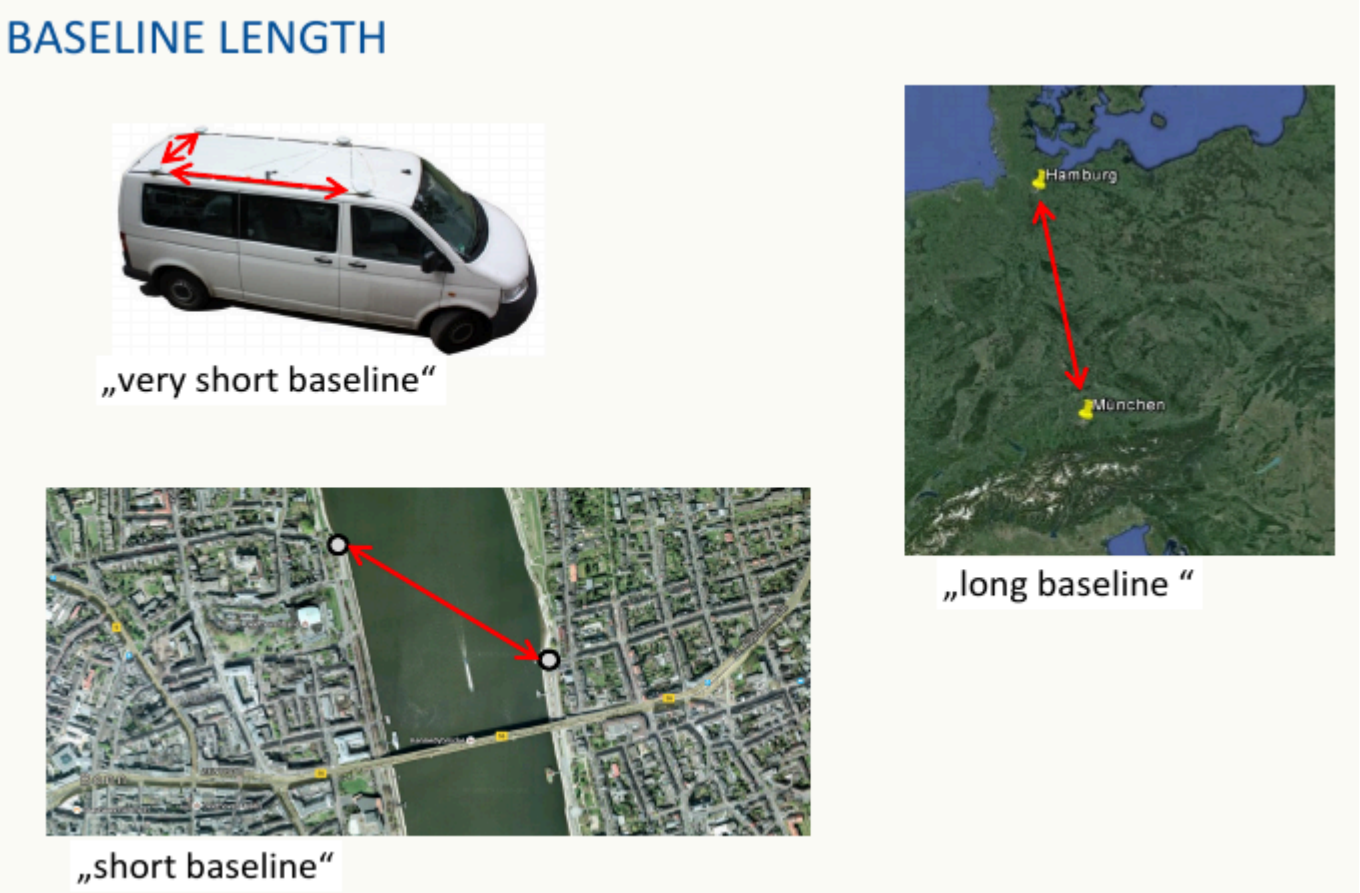


- Remaining errors after modelling of Ionosphere, Troposphere and satellite clock: **~ meters**
- Carrier phase measurement accuracy: **~mm**
- Target accuracy: **~cm**
- -> We need to remove systematic error better!
- -> Strategy: create **new observations** by building differences between avilable observations

Building Differences

IDEA:



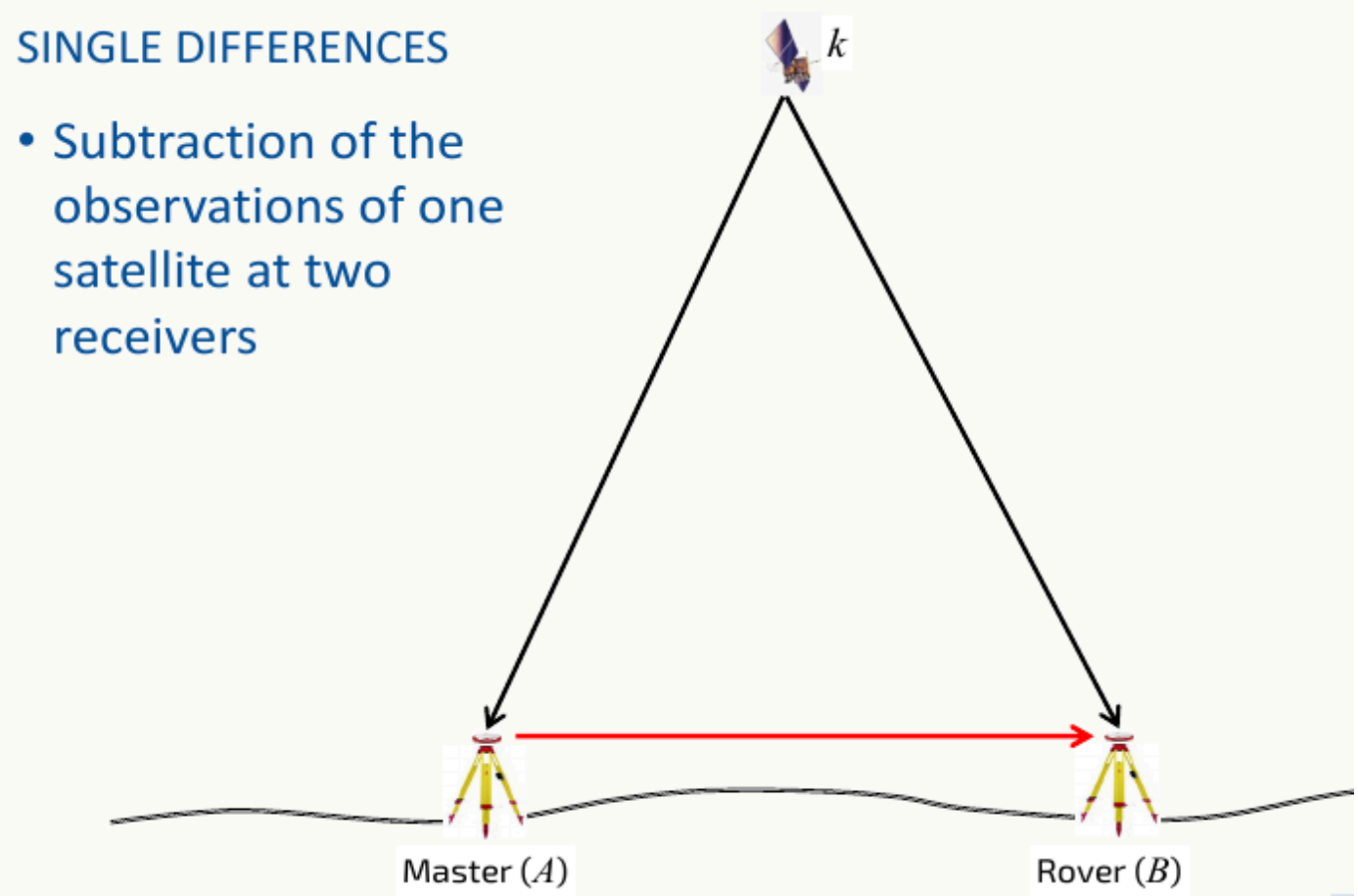


Carrier Phase Measurement Model:

$$\phi_A^k(t) = \frac{1}{\lambda}[r_A^k(t) + I_{\phi,A}^k(t) + T_A^k(t)] + \frac{c}{\lambda}[\delta t_A(t) - \delta t^k(t-\tau)] + N_A^k(t) + \varepsilon_A^\phi(t)$$

- A ... Station/Receiver Index (e.g. A: Master, B:Rover)

- k ... Satellite Index
- N_A^k ... integer phase ambiguity for satellite k in receiver A



SINGLE DIFFERENCES $\phi_{AB}^k(t) = \phi_A^k(t) - \phi_B^k(t)$

Subtraction

$$\phi_A^k(t) = 1/\lambda \left[r_A^k(t) + I_{\phi,A}^k(t) + T_A^k(t)\right] + c/\lambda \left[\delta t_A(t) - \delta t^k(t-\tau)\right] + N_A^k + \epsilon_\phi(t)$$

$$\phi_B^k(t) = 1/\lambda \left[r_B^k(t) + I_{\phi,B}^k(t) + T_B^k(t)\right] + c/\lambda \left[\delta t_B(t) - \delta t^k(t-\tau)\right] + N_B^k + \epsilon_\phi(t)$$

Atmospheric delay ~ the same for short base lines

Satellite clock

Satellite orbit errors are also minimized by subtraction!

BUILDING DIFFERENCES

WHAT IS THE INFLUENCE OF THE ORBIT ERROR ON THE BASELINE?

- Rule of thumb

➔ $$\boxed{\frac{db}{b} = \frac{dr}{R}}$$

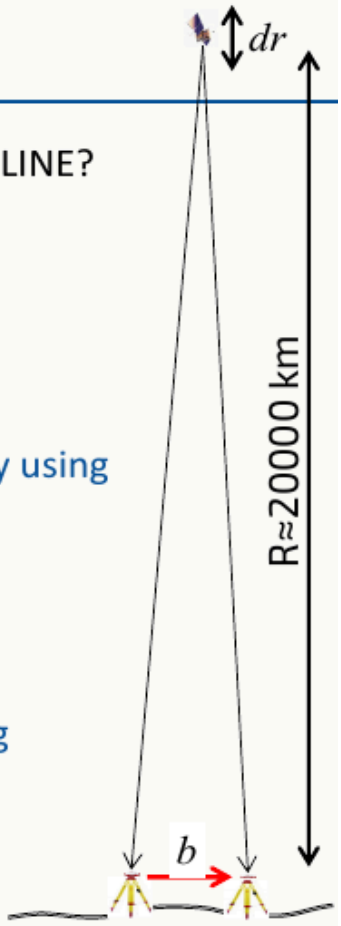
$$R \approx 20000 \text{ km}$$

- Example 1: Maximum baseline length to have 1cm accuracy using broadcast orbit data (2m accuracy)? (Ignoring other errors)

$$b = db \cdot \frac{R}{dr} = 0.01\,m \cdot \frac{20000\,km}{2\,m} = 100km$$

- Example 2: theoretical accuracy of a 1000km baseline using IGS precise orbits (5cm accuracy) (Ignoring other errors)

$$db = b \cdot \frac{dr}{R} = 1000\,km \cdot \frac{0.05\,m}{20000\,km} = 2{,}5mm$$



BUILDING DIFFERENCES

SINGLE DIFFERENCES (SD)
MEASUREMENT MODEL

$$r^k_{AB} = r^k_A - r^k_B$$
$$\delta t^k_{AB} = \delta t^k_A - \delta t^k_B$$
$$N^k_{AB} = N^k_A - N^k_B$$
$$\epsilon^k_{AB} = \epsilon^k_A - \epsilon^k_B$$

$$\phi^k_{AB}(t) = \phi^k_A(t) - \phi^k_B(t)$$
$$= \frac{1}{\lambda} r^k_{AB}(t) + \frac{c}{\lambda}\delta t_{AB}(t) + N^k_{AB}(t) + \epsilon^k_{AB}(t) \quad [cycles]$$

Baseline vector — SD-ambiguities

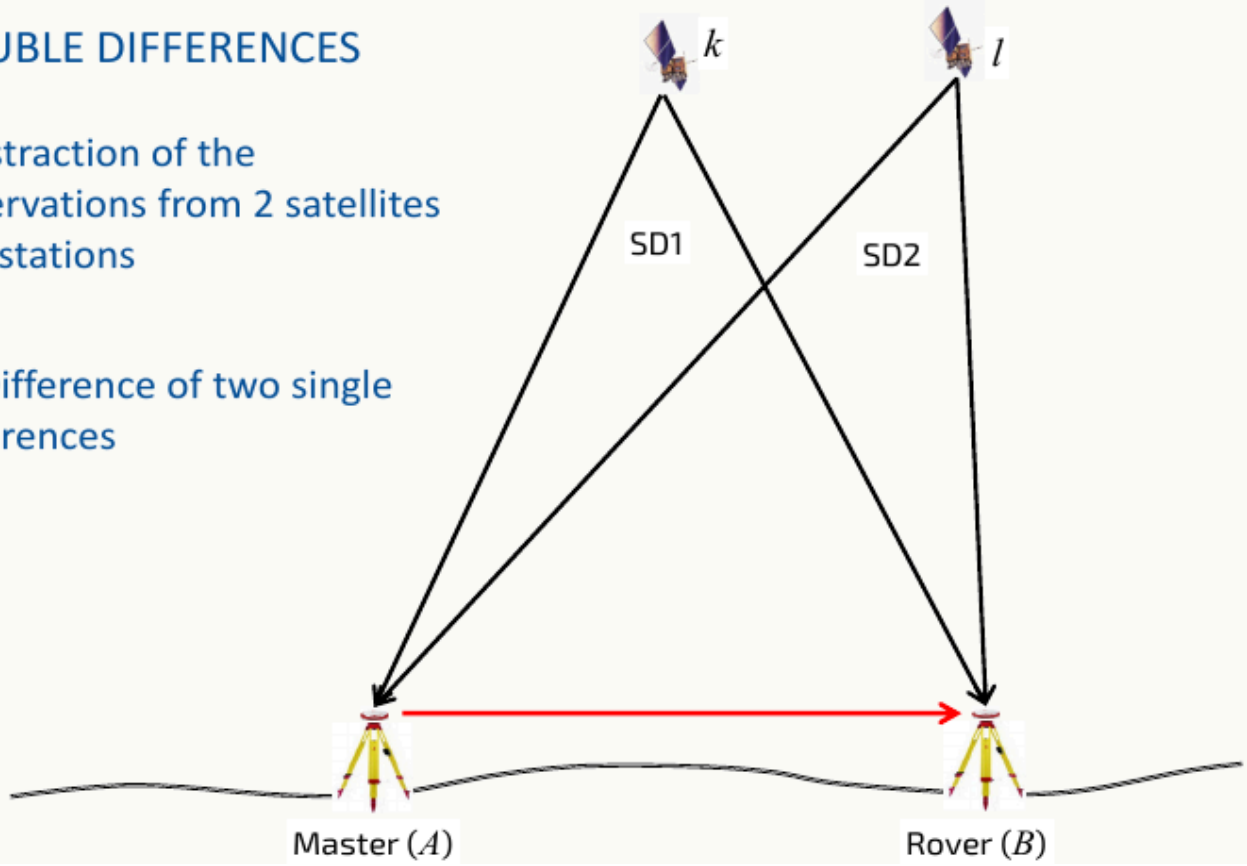
- Building differences increases noise by a factor of √2
- No reduction of multipath!

BUILDING DIFFERENCES

DOUBLE DIFFERENCES

Substraction of the observations from 2 satellites at 2 stations

=> Difference of two single differences



BUILDING DIFFERENCES

DOUBLE DIFFERENCES

$$\phi^{kl}_{AB}(t) = \phi^{k}_{AB}(t) - \phi^{l}_{AB}(t)$$

Subtraction

$$\phi^{k}_{AB}(t) = \frac{1}{\lambda} r^{k}_{AB}(t) + \cancel{\frac{c}{\lambda}\delta t_{AB}(t)} + N^{k}_{AB}(t) + \epsilon^{k}_{AB}(t)$$

receiver clock offset

SD-ambiguities

$$\phi^{l}_{AB}(t) = \frac{1}{\lambda} r^{l}_{AB}(t) + \cancel{\frac{c}{\lambda}\delta t_{AB}(t)} + N^{l}_{AB}(t) + \epsilon^{l}_{AB}(t)$$

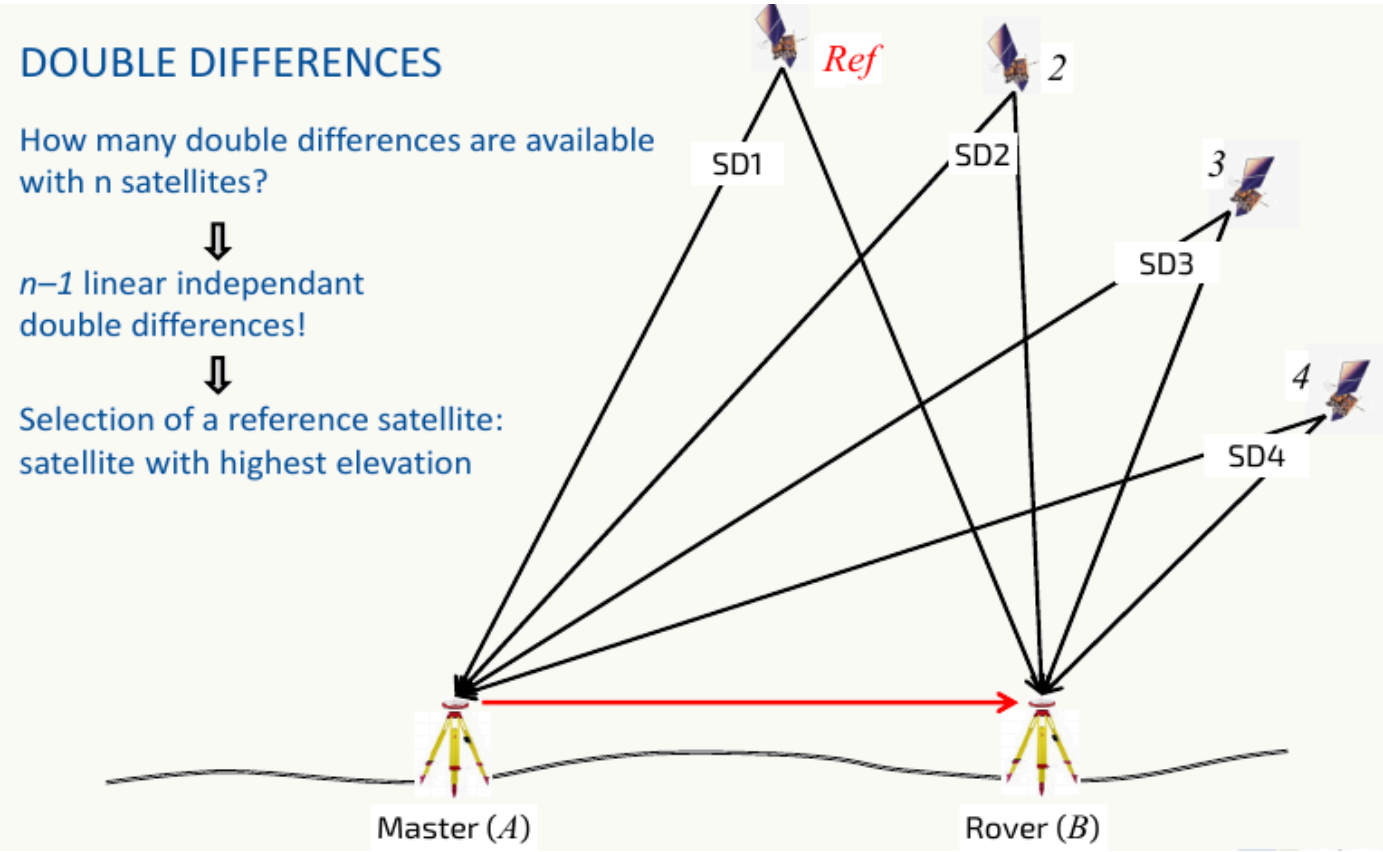
BUILDING DIFFERENCES

DOUBLE DIFFERENCES (DD) MEASUREMENT MODEL

$$r_{AB}^{kl} = r_{AB}^{k} - r_{AB}^{l}$$
$$N_{AB}^{kl} = N_{AB}^{k} - N_{AB}^{l}$$
$$\epsilon_{AB}^{kl} = \epsilon_{AB}^{k} - \epsilon_{AB}^{l}$$

$$\begin{aligned}\phi_{AB}^{kl}(t) &= \phi_{AB}^{k}(t) - \phi_{AB}^{l}(t) \\ &= \frac{1}{\lambda} r_{AB}^{kl}(t) + N_{AB}^{kl}(t) + \epsilon_{AB}^{kl}(t) \quad [cycles]\end{aligned}$$

- Noise multiplied again by a factor of $\sqrt{2}$
- Multipath is still not reduced



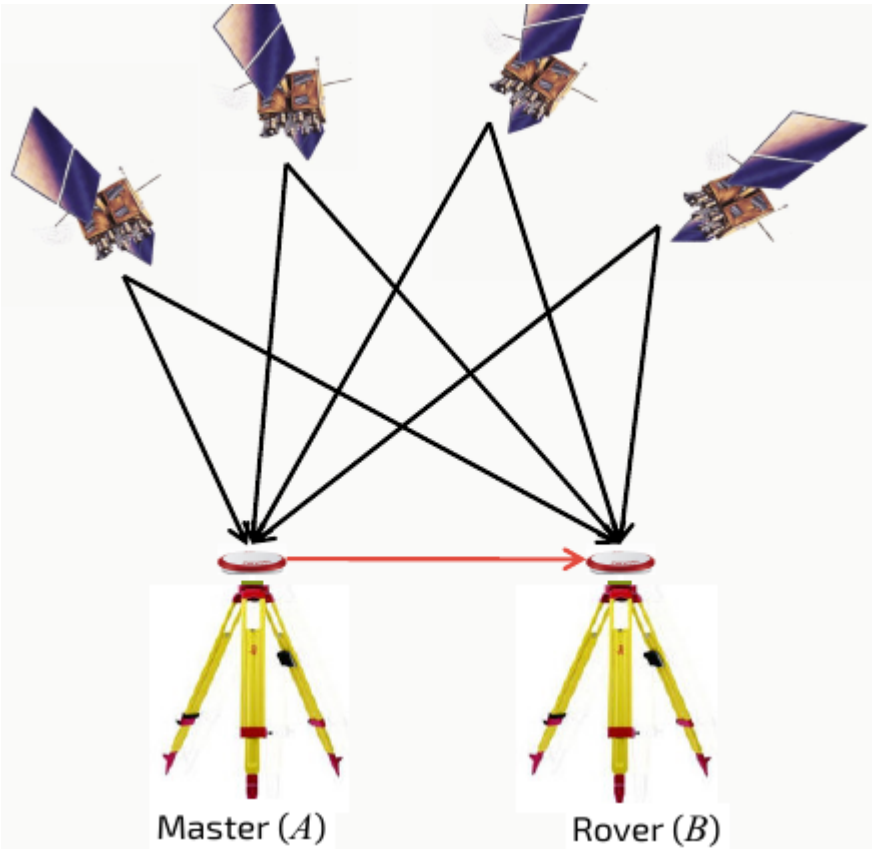
Summarise

- By building differences at two receivers, systematic errors can be reduced or eliminated
- **Single Difference (SD)**: Differences of received signals from one satellite at two receivers
  - Reduction - dependant on the length of the baseline
    - Troposperic refraction
    - Ionospheric refraction

        - Satellite clock
        - Satellite orbit errors (+ other satellite related errors)
- **Double Differences (DD):** Difference of two single differences from two satellites
    - Elimination of receiver clock errors
    - Reduction of receiver dependent errors (e.g. antenna)

Relative Positioning

- Observations of GPS signals at two station
- Position of the master station is known
- Position of the rover is not known
- **Goal**: determination of the vector between the two antennas
- Relative GPS position determination



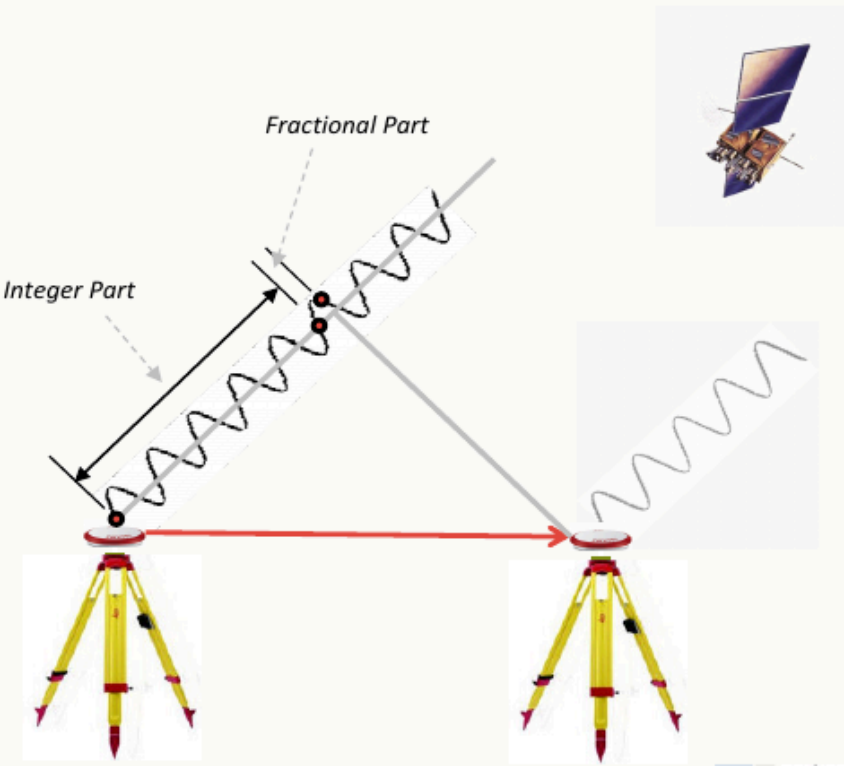
RELATIVE POSITIONING

Fractional Part Φ_{AB}:

➔ Phase difference of the signals observed at two receivers (Single Differences)

Integer Part:

➔ Unknown number of full wave cycles N·λ



- Wanted:

Amiguities N

*GEOMETRIC VISUALIZATION

SINGLE DIFFERENCES

Line-of-sight (LOS) vector $\mathbf{1}_B^{(k)}$

projects baseline vector $\mathbf{x}_{AB}$

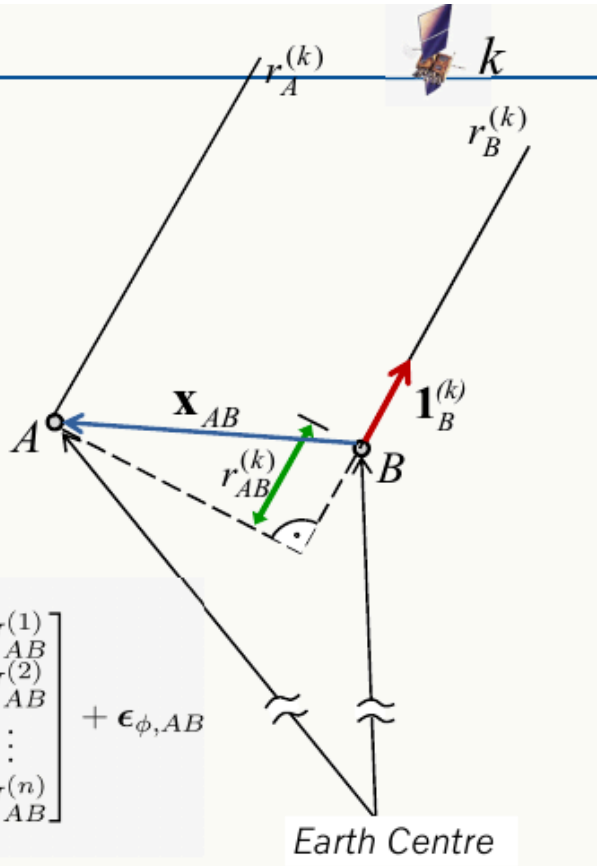
onto the single difference $r_{AB}^{(k)}$

$$r_{AB}^{(k)} = r_A^{(k)} - r_B^{(k)} = -\mathbf{1}_B^{(k)} \cdot \mathbf{x}_{AB}$$

Measurement model

$$\mathbf{\Phi} = \frac{1}{\lambda}\begin{bmatrix} \left(-\mathbf{1}_B^{(1)}\right)^T \\ \left(-\mathbf{1}_B^{(2)}\right)^T \\ \vdots \\ \left(-\mathbf{1}_B^{(n)}\right)^T \end{bmatrix} \cdot \mathbf{x}_{AB} + \frac{c}{\lambda}\delta t_{AB} \cdot \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + \begin{bmatrix} N_{AB}^{(1)} \\ N_{AB}^{(2)} \\ \vdots \\ N_{AB}^{(n)} \end{bmatrix} + \epsilon_{\phi,AB}$$

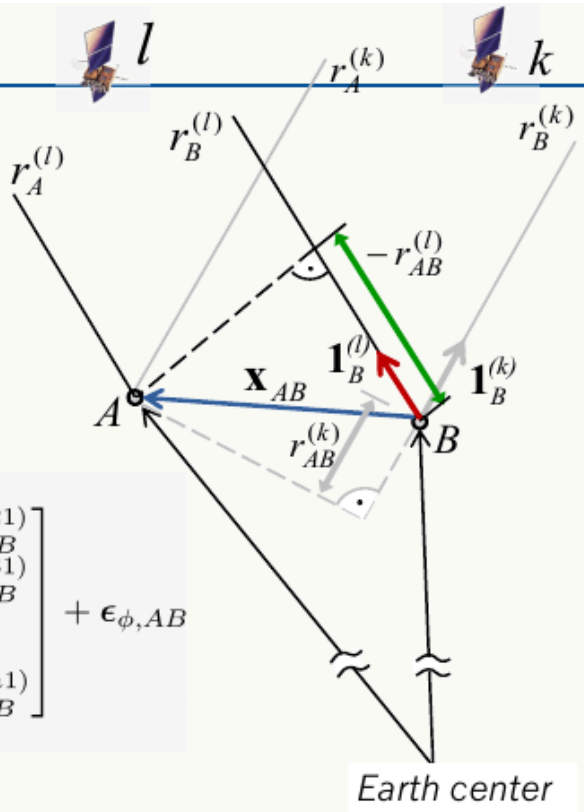
Vector of phase differences for all n satellites



*GEOMETRIC VISUALIZATION

DOUBLE DIFFERENCES

$$r_{AB}^{(kl)} = \left(r_A^{(k)} - r_B^{(k)}\right) - \boxed{\left(r_A^{(l)} - r_B^{(l)}\right)}$$
$$= -\left(\mathbf{1}_B^{(k)} - \boxed{\mathbf{1}_B^{(l)}}\right) \cdot \boxed{\mathbf{x}_{AB}}$$



Measurement model

$$\begin{bmatrix} \Phi_{AB}^{(21)} \\ \Phi_{AB}^{(31)} \\ \vdots \\ \Phi_{AB}^{(n1)} \end{bmatrix} = \frac{1}{\lambda} \underbrace{\begin{bmatrix} -\left(\mathbf{1}_B^{(2)} - \mathbf{1}_B^{(1)}\right)^T \\ -\left(\mathbf{1}_B^{(3)} - \mathbf{1}_B^{(1)}\right)^T \\ \vdots \\ -\left(\mathbf{1}_B^{(n)} - \mathbf{1}_B^{(1)}\right)^T \end{bmatrix}}_{r_{AB}^{(kl)}} \cdot \mathbf{x}_{AB} + \begin{bmatrix} N_{AB}^{(21)} \\ N_{AB}^{(31)} \\ \vdots \\ N_{AB}^{(n1)} \end{bmatrix} + \epsilon_{\phi,AB}$$

n-1 double differences, satellite 1 is reference

$$\begin{bmatrix} \lambda \cdot \Phi_q^{(21)} \\ \lambda \cdot \Phi_q^{(31)} \\ \vdots \\ \lambda \cdot \Phi_q^{(n1)} \end{bmatrix} = \begin{bmatrix} -\left(\mathbf{1}_B^{(2)} - \mathbf{1}_B^{(1)}\right)^T \\ -\left(\mathbf{1}_B^{(3)} - \mathbf{1}_B^{(1)}\right)^T \\ \vdots \\ -\left(\mathbf{1}_B^{(n)} - \mathbf{1}_B^{(1)}\right)^T \end{bmatrix} \cdot \mathbf{x}_{AB} + \lambda \cdot \begin{bmatrix} N_q^{(21)} \\ N_q^{(31)} \\ \vdots \\ N_q^{(n1)} \end{bmatrix} + \epsilon_{\phi,q} \quad [m]$$

Subscript q: frequency indicator (e.g. L1 or L2 frequency observations)

multiplication with λ leads to [m]

- Ambiguities are constant, as long as there is no signal interruption!
- Unknown parameter: n-1 DD ambiguities, baseline vector $\boldsymbol{x}_{AB}$
- Estimation via non linear least squares (similar to single point positioning)

NON-LINEAR LEAST SQUARES

- Initial value for rover $\mathbf{x}_{B,0}$ ➔ initial distance rover – satellite $r_{B,0}^{(k)}$

$$r_{B,0}^{(k)} = \sqrt{(x^{(k)} - x_{B,0})^2 + (y^{(k)} - y_{B,0})^2 + (z^{(k)} - z_{B,0})^2} = \|\mathbf{x}^{(k)} - \mathbf{x}_{B,0}\|$$

- Known master coordinates ➔distance master – satellite $r_A^{(k)}$

$$r_A^{(k)} = \sqrt{(x^{(k)} - x_A)^2 + (y^{(k)} - y_A)^2 + (z^{(k)} - z_A)^2} = \|\mathbf{x}^{(k)} - \mathbf{x}_A\|$$

- Observation vector within non-linear least squares

$$\Delta l_q^{(kl)}(t_1) = \lambda \cdot \phi_q^{(kl)} - r_{B,0}^{(l)}(t_1) + r_{B,0}^{(k)}(t_1) + r_A^{(l)}(t_1) - r_A^{(k)}(t_1)$$

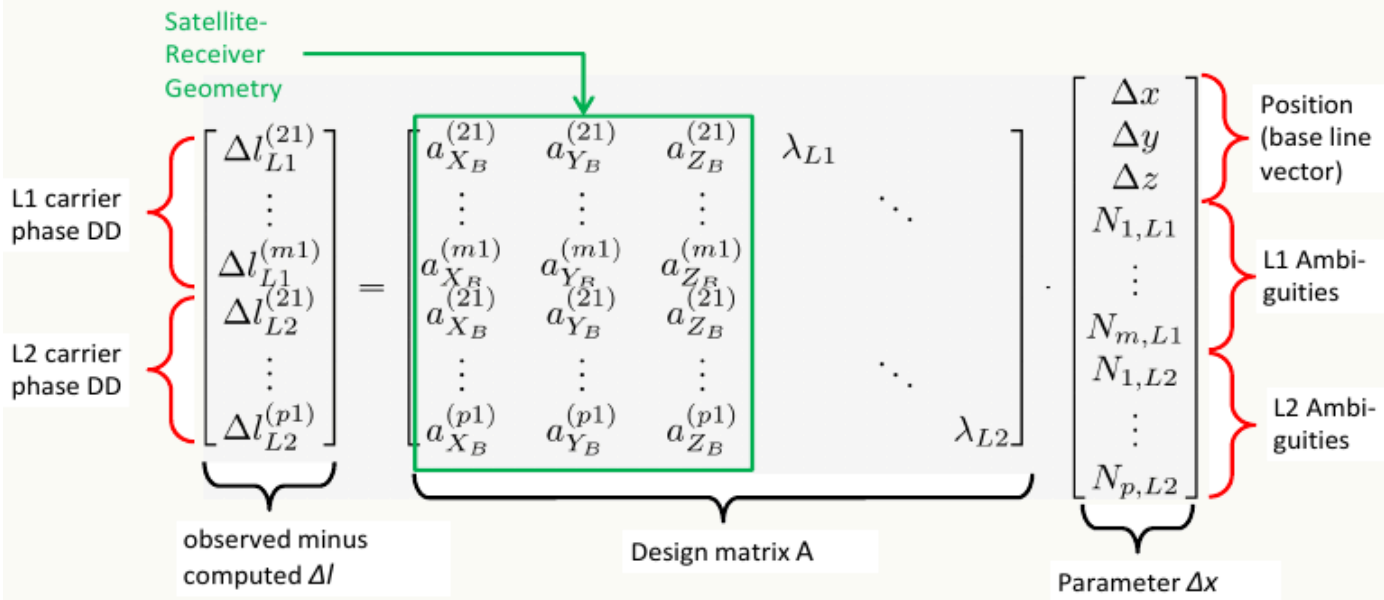
- Designmatrix A

$$a_{X_B}^{(kl)}(t_1) = -\frac{x^{(l)}(t_1) - x_{B,0}}{r_{B,0}^{(l)}} + \frac{x^{(k)}(t_1) - x_{B,0}}{r_{B,0}^{(k)}} \qquad a_{Y_B}^{(kl)}(t_1) = -\frac{y^{(l)}(t_1) - y_{B,0}}{r_{B,0}^{(l)}} + \frac{y^{(k)}(t_1) - y_{B,0}}{r_{B,0}^{(k)}}$$

$$a_{Z_B}^{(kl)}(t_1) = -\frac{z^{(l)}(t_1) - z_{B,0}}{r_{B,0}^{(l)}} + \frac{z^{(k)}(t_1) - z_{B,0}}{r_{B,0}^{(k)}}$$

NON LINEAR LEAST SQUARES

Number of observations for n satellites:
m=n-1 carrier phase DD on L1 + p carrier phase DD on L2

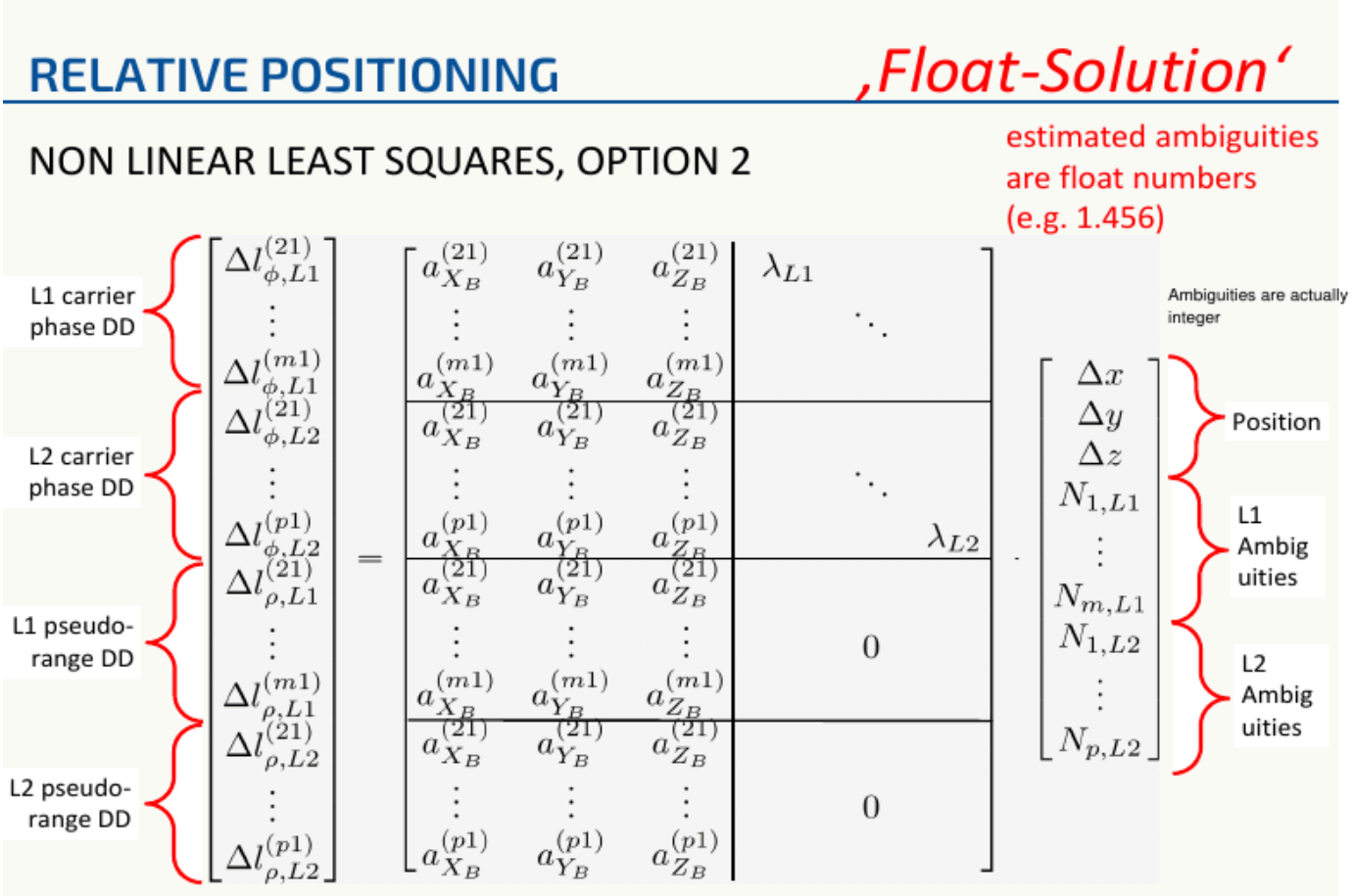


Non Linear Least Squares

- Problem: underdetermined system
  - 2(n-1) observations <--> 2(n-1)+3 unknown parameters per epoch
- Option 1: Use data from multiple epochs
  - No fast ambiguity resolution within a single epoch

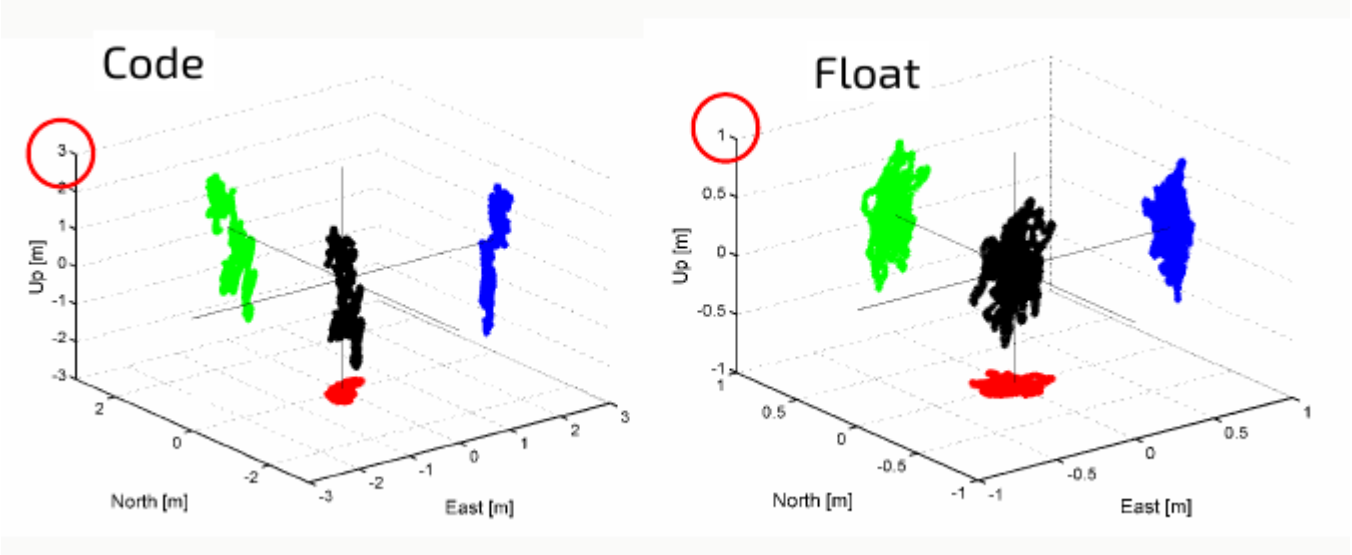
- Option 2: Use pseudo-range (code observations)
  - Doubling the number of observations
  - But: observations are less accurate -> ambiguity estimation is less accurate
  - Ambiguity resolution in a single epoch is possible
  - Double difference of pseudoranges:
    - $\rho_{AB}^{(kl)}(t) = r_{AB}^{(kl)}(t) + \varepsilon_{\rho,AB}^{(kl)}(t)$ in [m]

In case of Option 2

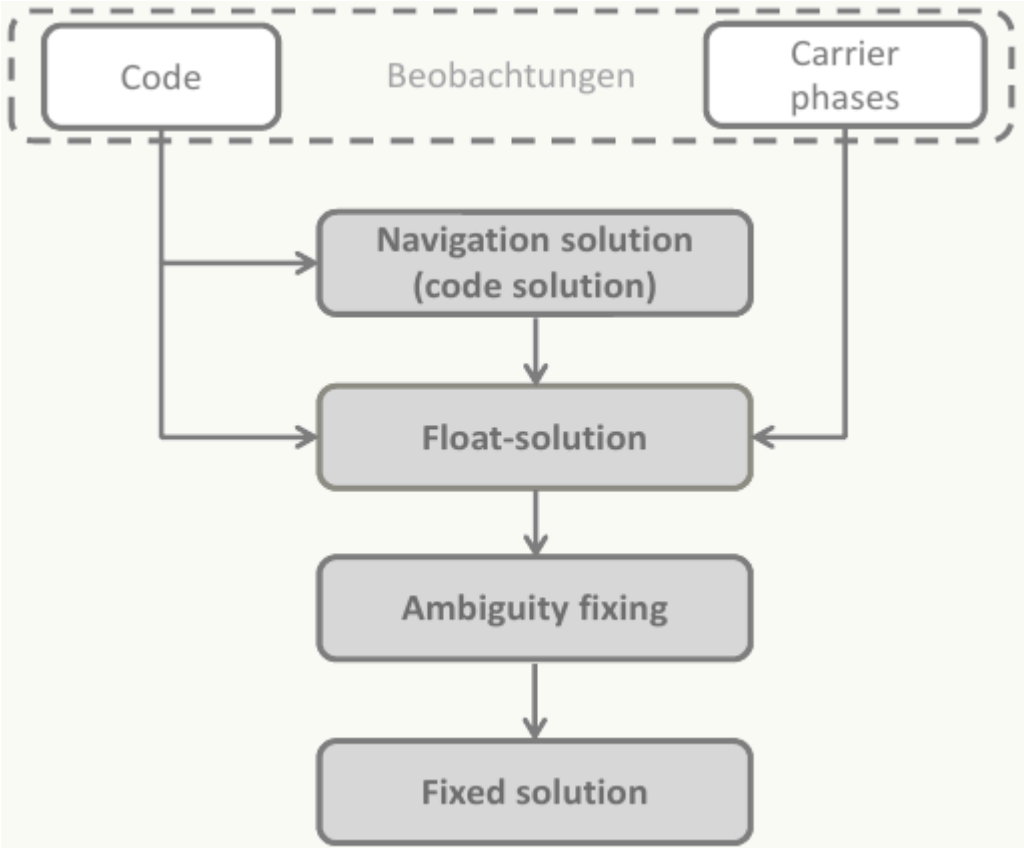


Code solution vs Float Solution

Precision of float solution is higher than code solution

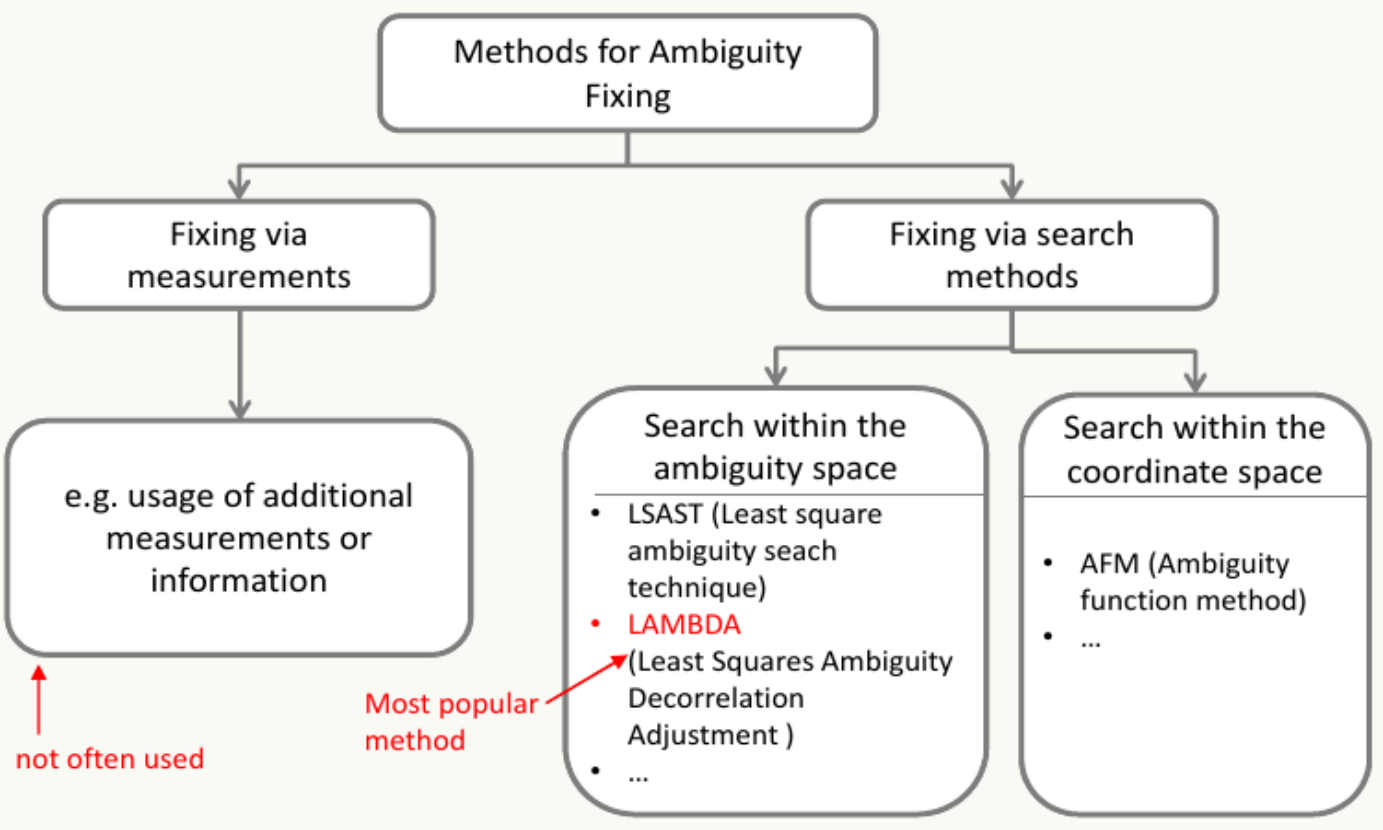


But: Fixing the float values ambiguities to integer numbers is the goal!



Ambiguity Fixing

Ambiguities can not be fixed by simple rounding
-> correct integer values have to be found by other methods



Ambiguity Function Method

- Assume correct position within cube around float solution of the rover
- Define **cube as search space**
- **Calculate DD** observations for **all candidate** positions
- Test candidate positions with evaluation function ARF
  - -> Difference between observed and calculated DD is integer ambiguity, if candidate position is correct
  - The more integer, the more correct
- using $cos(2\pi x)$
- Disadvantages: Often no unambiguous solution and many candidates to test

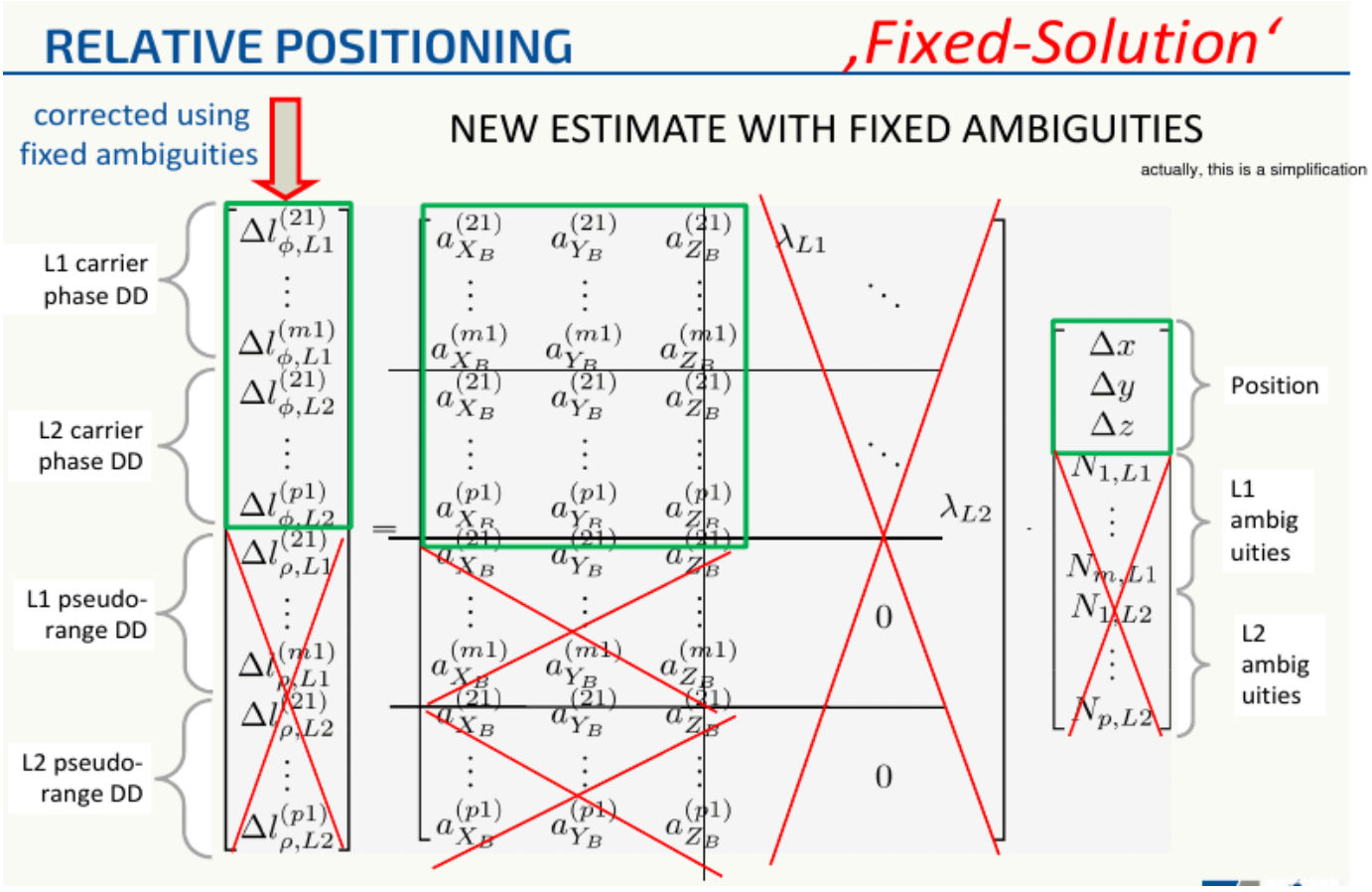
Lambda Method

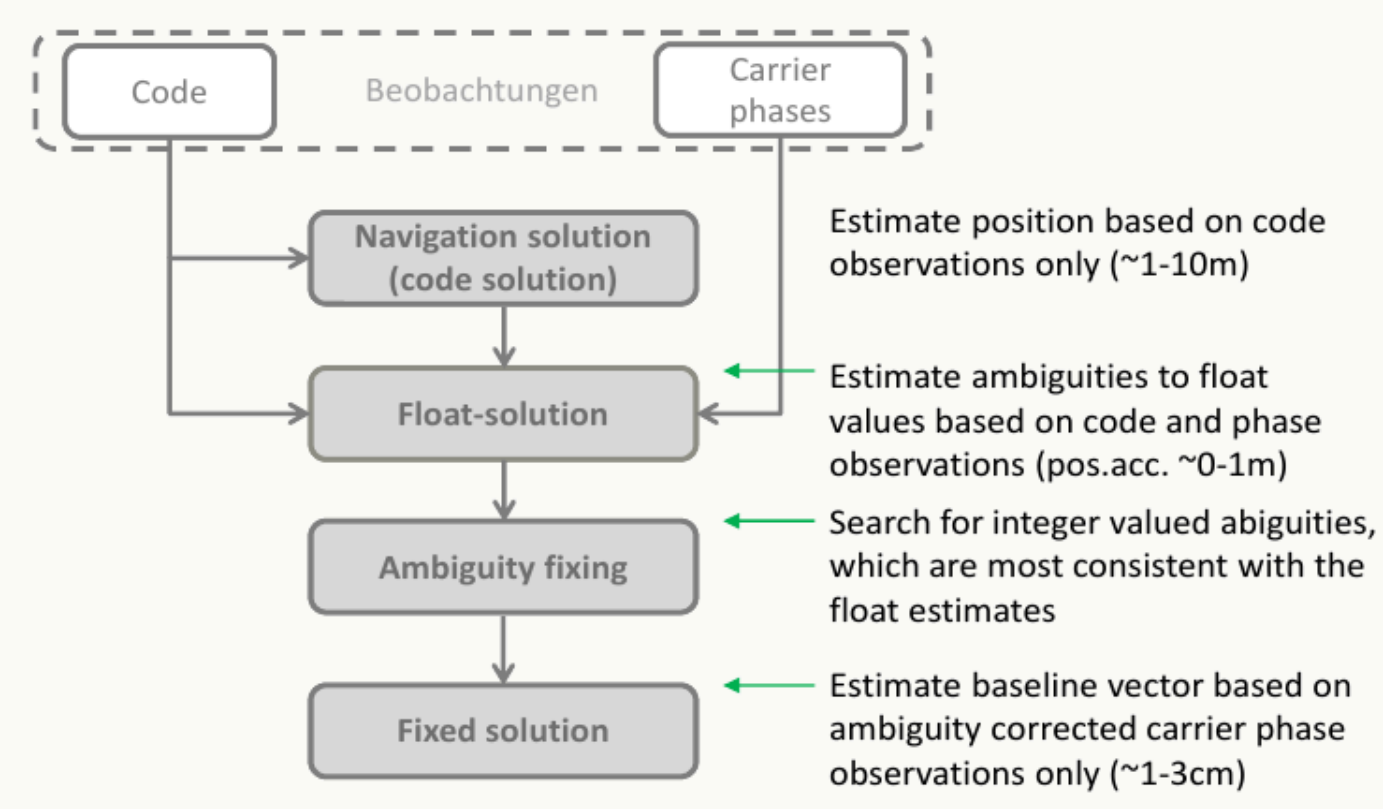
- Float Solution estimates DD ambiguities as float values, but true ambiguities are integer values
- Differences between float and integer ambiguities should be minimal -> Integer least squares estimation
- -> **Solution is a search problem**
- Search Space: Hyperellipsoid
  - center around Float ambiguities
  - Form is according to float estimations covariance
  - Problem: Correlations

Successful Ambiguity Fixing depends on

- Quality of the float solution. This depends on
  - Baseline length (reduction of atmospheric refraction effect)
  - Satellite geometry
  - Number of used frequencies
  - Accuracy of code and phase observations (measurement conditions, multipath,...)
- What helps
  - Long observation time
  - Change of satellite geometry
  - Free horizon

Fixed - Solution




Code
Beobachtungen
Carrier phases
Navigation solution (code solution)
Float-solution
Ambiguity fixing
Fixed solution
Estimate position based on code observations only (~1-10m)
Estimate ambiguities to float values based on code and phase observations (pos.acc. ~0-1m)
Search for integer valued abiguities, which are most consistent with the float estimates
Estimate baseline vector based on ambiguity corrected carrier phase observations only (~1-3cm)